

# Rosetta

**A text-to-SQL system that refuses to answer questions it cannot answer honestly — and measures how often guessing produced a confident wrong number.**

Project report, written from first principles. No prior knowledge of text-to-SQL or Indian public finance is assumed.

Commit `f0a8732` · 14 September 2026 · every figure measured and reproducible from this repository.

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## How to read this

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Parts I and II are background: what the problem is and what the data means. Nothing project-specific until Part III. If you already know what text-to-SQL is, start at Part III.

Part VI is the section worth reading twice. It records the design decisions that turned out to be **wrong**, and what replaced them. In a project whose subject is confidently wrong numbers, the mistakes are the most informative part of the record.

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## Part I — Foundations

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### 1. What a database query is

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Information in a database sits in tables — rows and columns, like a spreadsheet with rules. To get anything out you write **SQL**:

```
SELECT SUM(be_amount_crore)
FROM budget_allocations
WHERE fiscal_year = '2019-20'
```

Read it as: *add up the `be_amount_crore` column, across every row where the year is 2019-20.*

SQL takes weeks to learn. Learning a *particular organisation's* database takes months — not because the syntax is hard, but because you have to learn what the columns mean, which ones are safe to add together, and which of the six columns with "amount" in the name is the one people actually quote.

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### 2. What text-to-SQL is

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**Text-to-SQL** removes that barrier: you ask in English, a language model writes the SQL, the database answers.

It is one of the most commercially attractive uses of language models, for a simple reason — every organisation has a database, and almost nobody in the organisation can query it. The people who want the numbers and the people who can extract them are different people, and the gap between them costs real time.

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### 3. Why it works in demos and fails in production

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Text-to-SQL demos are reliably impressive. Production deployments are reliably disappointing. The gap is measurable, and the measurement is stark.

Published benchmark results show **the same model scoring 91.2% on Spider 1.0 and 21.3% on Spider 2.0** — a 70-point collapse on what is nominally the same task. On **BEAVER**, a benchmark built from real enterprise warehouse data, off-the-shelf models approach roughly **0%**.

The model did not get worse between those benchmarks. The databases got realistic.

|                      | SPIDER 1.0 | SPIDER 2.0                                                               |
|----------------------|------------|--------------------------------------------------------------------------|
| Tables               | a handful  | many                                                                     |
| Columns per database | tens       | <b>~800 average</b>                                                      |
| Column names         | tidy       | <code>revenue</code> , <code>revenue_net</code> , <code>rev_final</code> |
| Join paths           | obvious    | undocumented, tribal knowledge                                           |
| Permissions          | superuser  | ignored by the benchmark                                                 |

The documented causes are all about **context**, not capability:

- Ambiguous column names
- Undocumented join paths that exist only in someone's head
- Access controls benchmarks skip by running as superuser
- Dialect differences between database engines

The consensus prescription in the literature is **"fix the context, not the model."** This project takes that seriously and then tries to measure whether it is true.

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## 4. The failure that actually matters

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Almost every text-to-SQL evaluation reports one number: **execution accuracy** — the share of questions where the generated query returned the right rows.

That metric treats these two outcomes as equally bad:

|                          |                                       |
|--------------------------|---------------------------------------|
| the query errored        | the user sees an error and asks again |
| the query returned 8,412 | the user puts 8,412 in a report       |

They are not equally bad.

An error is **visible**. It announces itself, costs a retry, and nobody acts on it. A clean wrong number is **invisible** — and it propagates, into a slide, a board pack, a press release. Nobody re-checks a figure that arrived without an error attached.

**This project's headline metric is the confidently-wrong rate:** the share of questions answered with a clean, plausible, *wrong* figure. It is rarely measured, which is exactly why it is worth measuring.

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## Part II — The domain

The database models the **Indian Union Budget**. You do not need to know Indian public finance to follow this report, but six conventions matter, because the entire system is built around them.

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## 5. Budget Estimate, Revised Estimate, Actuals

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Every fiscal year has three different numbers for the same thing.

| FIGURE                       | WHAT IT IS                                                            | WHEN IT EXISTS               |
|------------------------------|-----------------------------------------------------------------------|------------------------------|
| <b>Budget Estimate</b> (BE)  | What was proposed when the budget was presented, in February          | Every year                   |
| <b>Revised Estimate</b> (RE) | The mid-year correction, published with the <i>next</i> year's budget | All but the newest year      |
| <b>Actuals</b>               | What was really spent, once the accounts close                        | Roughly two years in arrears |

They differ routinely by 10–20%. So "*the budget for the Ministry of Railways in 2020-21*" is not one number. It is three, and the question does not say which.

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## 6. Crore and lakh

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Indian numbering uses **lakh** (100,000) and **crore** (10,000,000). One crore = 100 lakh. Budget documents mix them freely.

If a query sums a column measured in lakh and reports it as crore, the answer is **100× too large** — and still looks like a plausible budget figure.

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## 7. Plan and Non-Plan, abolished in FY2017-18

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Until 2016-17, Indian government spending was classified as "Plan" or "Non-Plan". From 2017-18 that classification was **abolished** and replaced by Scheme/Establishment.

A question asking for Plan expenditure in 2019-20 therefore has no answer. Not zero — *undefined*. The distinction matters enormously, because a query that returns zero looks like an answer.

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## 8. Ministries are renamed and merged

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- Ministry of Human Resource Development → Department of School Education and Literacy (2020)
- Ministry of Water Resources → merged into Jal Shakti (2019)
- Urban Development + Housing & Urban Poverty Alleviation → Housing and Urban Affairs (2017)

The **money** is continuous. The **name** is not. Filtering on one name across a decade produces a spending cliff that never happened.

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## 9. Not every budget line is a ministry

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The published budget lists **Interest Payments**, **Defence Pensions** and **Transfers to States** alongside actual ministries. These are real spending — about **44% of the total** — but they are not administrative bodies.

"Total expenditure" includes them. "What ministries spent" does not. On this data the two differ by **5.2×**.

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## 10. The budget is organised by department, not ministry

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There is no "Ministry of Education" line in the published budget. Its money appears under *Department of School Education and Literacy* and *Department of Higher Education*.

This was discovered while building the loader, not assumed — and it is why a question about "the education ministry" needs a certified reading rather than a guess.

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## Part III — The research that informed the design

Four findings from the literature shaped this project. Each is load-bearing: remove it and a design decision downstream stops making sense.

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### 11. The benchmark-to-warehouse collapse

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The 91.2% → 21.3% Spider 1.0/2.0 gap, and BEAVER's ~0%, establish that **the problem is not model capability**. This is the premise of the whole project. It is why the intervention is a semantic layer rather than a better model or a fine-tune.

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### 12. "Fix the context, not the model"

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The consensus remedy in the text-to-SQL literature is to repair the *context* the model is given: document what columns mean, certify metric definitions, write down valid join paths.

This directly produced the `semantic/` directory (Part IV) and the decision to make the ablation switch that directory on and off rather than swap models.

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### 13. The semantic layer as an established pattern

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A "semantic layer" is not a novel invention here — it is a standard component in business-intelligence tooling (dbt metrics, LookML, Cube), where its purpose is to stop two analysts computing "revenue" two different ways.

The contribution of this project is not inventing it, but **measuring what it is worth** when placed in front of a language model, on a schema deliberately built to punish its absence.

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### 14. Selective prediction and abstention

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In machine learning, **selective prediction** is the idea that a model may decline to predict when confidence is low, trading coverage for accuracy. Abstention is well studied in classification.

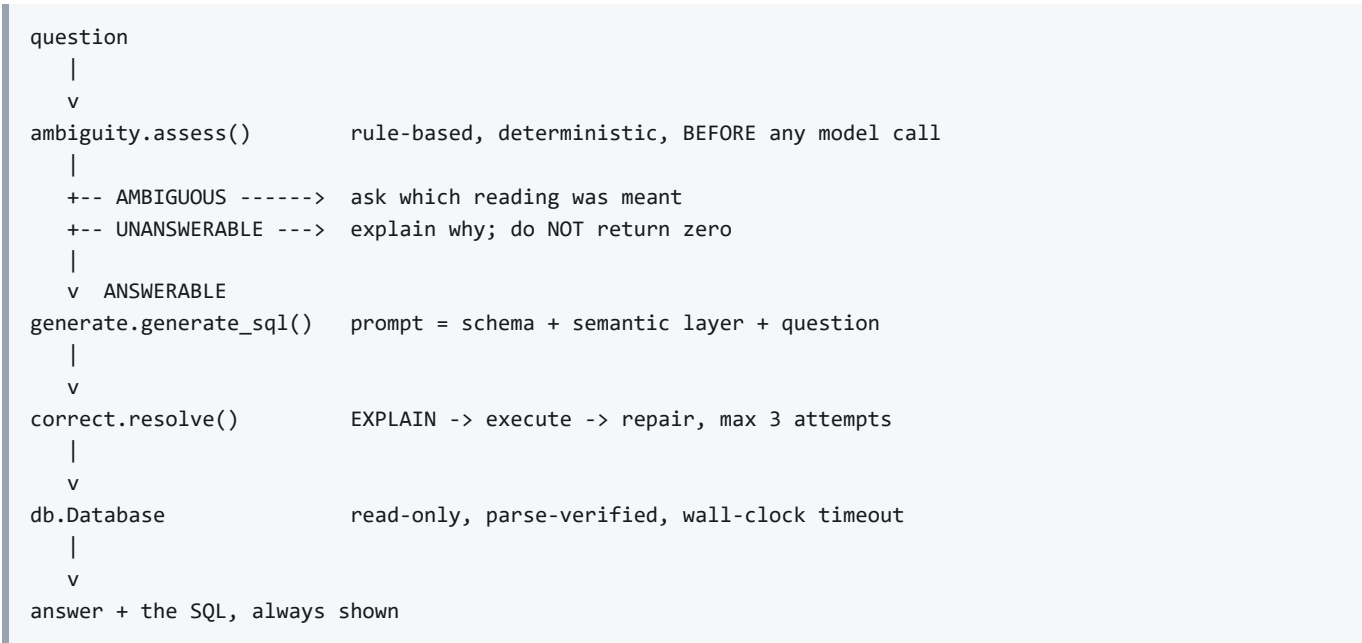
The equivalent for text-to-SQL is under-explored, and it motivated the central design choice here: an under-specified question should produce a *question back*, not a number. Part V.3 explains why that decision is made by rules rather than by the model.

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## Part IV — Architecture, module by module

The system is roughly 6,000 lines of Python across `src/rosetta/` (the library) and `scripts/` (entry points), plus three YAML files that hold the domain knowledge.

## 15. Data flow



## 16. The library

| MODULE                    | RESPONSIBILITY                                                                                                                                                                                                                      |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>db.py</code>        | The execution sandbox. Nothing else touches the database. Guarantees read-only, SELECT-only (by parsing, not keyword matching), a wall-clock ceiling per query, and blocks 25 filesystem functions. Also does schema introspection. |
| <code>semantic.py</code>  | Loads the three YAML files, renders the parts that bear on ambiguity into prompt text, reads exact column value-domains live from the database, and resolves entity lineage across renames by graph traversal.                      |
| <code>ambiguity.py</code> | The refusal rules. Classifies a question as ANSWERABLE / AMBIGUOUS / UNANSWERABLE and, under the DISCLOSE policy, converts defaultable ambiguities into stated assumptions. Pure functions, no model, no database.                  |
| <code>generate.py</code>  | Prompt assembly and model calls. Defines a <code>Provider</code> protocol so a stub can replace the model in tests. Extracts a single SQL statement from whatever prose the model wrapped it in.                                    |
| <code>correct.py</code>   | The repair loop. Generate → EXPLAIN → execute → feed the error back, capped at three attempts. Records whether repair rescued each query.                                                                                           |
| <code>pipeline.py</code>  | Orchestration. Holds the two switches that define the experiment: <code>use_semantic_layer</code> and <code>policy</code> .                                                                                                         |

| MODULE                   | RESPONSIBILITY                                                                                                                  |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <code>compare.py</code>  | Result-set comparison. Canonicalises every cell, compares rows as a multiset, decides whether row order was actually requested. |
| <code>evaluate.py</code> | The outcome taxonomy and the metrics. Turns one pipeline answer into one of six labelled outcomes.                              |

## 17. The domain knowledge

| FILE                                | HOLDS                                                                                                  |
|-------------------------------------|--------------------------------------------------------------------------------------------------------|
| <code>semantic/glossary.yaml</code> | What each ambiguous column means, its unit, and the specific mistake it invites                        |
| <code>semantic/metrics.yaml</code>  | Certified metric definitions, and which dimensions a question must pin before a metric can be computed |
| <code>semantic/joins.yaml</code>    | Join paths classified <b>certified</b> / <b>ambiguous</b> / <b>forbidden</b>                           |
| <code>eval/questions.yaml</code>    | The 60-question evaluation set with gold SQL                                                           |

## 18. The entry points

| SCRIPT                                   | PURPOSE                                                            |
|------------------------------------------|--------------------------------------------------------------------|
| <code>scripts/load_data.py</code>        | Builds the database, deterministically, from one seed              |
| <code>scripts/run_eval.py</code>         | Runs the ablation arms and prints the comparison                   |
| <code>scripts/verify_gold.py</code>      | Checks gold SQL, refusal-rule agreement, and all six data traps    |
| <code>scripts/audit_gold.py</code>       | Independently re-verifies all 32 gold queries by a different route |
| <code>scripts/adjudicate.py</code>       | Evidence behind the 28 refusal categorisations                     |
| <code>scripts/measure_variance.py</code> | How much of the result is harness noise                            |
| <code>scripts/demo.py</code>             | The side-by-side contrast on one ambiguous question                |
| <code>scripts/ask.py</code>              | Interactive querying                                               |

# Part V — Design decisions and why

## 19. Why a synthetic-detail database rather than real data

**Decision.** Entity names, entity count and every spending magnitude come from the published Union Budget. The row-level detail beneath them — scheme, account-head, state and month rows — is generated, normalised so it sums back to the real figures.

**Reasoning.** The real published file is entity-level only; it has no scheme or monthly breakdown. Building the traps required detail that does not exist publicly. The trade buys a repository that rebuilds from one command with no

external download, at the cost of not being able to claim the figures are real.

**Honesty requirement.** This must be stated wherever the project is described. "Real structure and magnitudes, synthetic detail" is defensible; letting a reader assume otherwise is not.

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## 20. Why deliberately ambiguous columns

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A schema that is merely *large* does not test anything interesting. Six ambiguities were designed in, each modelled on a real property of the published data, and each with the same shape: **a query that runs cleanly and returns a plausible wrong number.**

| TRAP            | THE MISTAKE                                                                 | COST                     |
|-----------------|-----------------------------------------------------------------------------|--------------------------|
| Units           | Summing <code>be_amount_lakh</code> instead of <code>be_amount_crore</code> | 100× too high            |
| Estimate basis  | Answering with BE when the question meant Actuals                           | 10–20% out               |
| Plan era        | Summing <code>plan_amount_crore</code> across all ten years, NULL for seven | 7 years silently dropped |
| Entity scope    | Counting functional lines as ministries, or excluding them from a total     | 5.2× apart               |
| Renames         | Filtering one ministry name across a rename boundary                        | series truncated         |
| Grain / fan-out | Joining annual to monthly without aggregating first                         | 33× inflated             |

All six are re-verified on every build by `verify_gold.py`, so those figures are measured rather than asserted.

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## 21. Why refusal is rule-based, not model-based

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This looks backwards at first — the system uses a language model for the hard part (writing SQL) and hand-written rules for the subtle part (deciding whether a question is answerable). Three reasons:

1. **Determinism.** If refusal came from a model, the headline number would move between runs for reasons unrelated to what is being measured. Measured effect: the semantic arm has **0.0 points of spread** across three runs; the baseline has 5.0.
2. **Auditability.** Every refusal traces to a named rule, so it can be defended to the person who got refused.
3. **Cost.** It runs before generation, so it is free when it fires. 28 of 60 questions never reach the model.

The underlying principle: **the decision not to answer is not delegated to the thing being guarded against.**

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## 22. Why EXPLAIN before execute

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`EXPLAIN` asks the database to *plan* a query without running it. It catches unknown columns, bad joins and type errors for the price of a plan, before the query spends its execution budget.

On a 182-column schema most first attempts fail on a column name, so catching that cheaply is most of the loop's value.

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## 23. Why multiset comparison

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Two result sets can be "the same answer" while differing in a dozen boring ways — column order, row order, integers as floats, NULL vs NaN, whitespace. Every cell is canonicalised first.

Rows are then compared as a **multiset** — a bag that counts duplicates — not a set. A set would let a query that lost its `GROUP BY` compare equal to one that did not.

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## 24. Why six outcomes instead of one accuracy number

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Collapsing everything into execution accuracy hides the distinction the project exists to measure. The taxonomy is:

| OUTCOME                   | MEANING                                                                  |
|---------------------------|--------------------------------------------------------------------------|
| Correct                   | Answered, result matches gold                                            |
| Confidently wrong         | Clean, plausible, wrong number ← <b>the headline</b>                     |
| Answered + disclosed      | Answered an under-specified question using a stated default, and said so |
| Wrong order               | Right values, wrong sequence                                             |
| Errored                   | Failed after repairs — visible, therefore safe                           |
| Refused rightly / wrongly | Declined correctly / declined something answerable                       |

Two deliberate choices inside it:

- **Wrong order is not confidently wrong.** A reader who gets correct rows in a different sequence has not been misled about any figure. Folding it in would inflate the result in the project's own favour.
- **Refused-wrongly is tracked on purpose.** A system that refuses everything scores zero confidently-wrong while being useless. The pair must be read together.

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## 25. Why local inference

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The eval loop makes several hundred model calls per pass and is run many times. A rate-limited API tier would throttle every run. Local inference via Ollama removes rate limits, needs no key, and costs nothing — at the price of lower absolute SQL quality, which is acceptable because the project measures a *delta*, not an absolute.

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## 26. Why the ablation holds the schema identical

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The two arms differ by exactly one argument. Schema text, instructions, question and repair context are byte-identical between them. Without that discipline, any measured difference could be attributed to prompt length rather than to the semantic layer.

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# Part VI — Decisions that turned out wrong

This is the most useful section in the report. Everything here was believed correct when written, and was changed after evidence contradicted it.

Three failure *shapes* recur. They are worth naming, because they generalise well beyond this project.

**Shape A — Agreement is not correctness.** Two artefacts written by the same author from the same assumption will agree with each other while both being wrong.

**Shape B — A rule that is right for the wrong reason looks identical to one that is right.** Until something removes the cover.

**Shape C — A number that moves is not a number.** Any improvement inside the measurement noise is not an improvement.

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## 27. Excluding functional lines from the entity list — *wrong*

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**Decision.** Treat only ministries and departments as spending entities; exclude Interest Payments, Defence Pensions, Transfers to States.

**Why it was wrong.** Those lines are ~44% of the budget. Excluding them made the database total ₹23.7 lakh crore against a real ₹47.7 lakh crore — the provenance claim that magnitudes are real stopped holding.

**Replaced by.** Include them, tagged with `entity_kind`. This turned a bug into a feature: "total expenditure" vs "what ministries spent" became a genuine certified-reading problem worth 5.2×.

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## 28. Leaving `department_id` unpopulated — *wrong*

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**Decision.** Leave the column NULL; nothing needed it.

**Why it was wrong.** A 100%-NULL foreign key means any join through it returns nothing, and `SUM` over nothing returns `NULL` — which reads as a clean answer. The model joined through it and produced exactly that. The failure was real but caused by a dangling column rather than by any designed ambiguity, so it would have dominated the headline for the wrong reason.

**Replaced by.** Populated it. Both join paths now agree.

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## 29. A ministry keyword that matched nothing — *wrong*

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**Decision.** Attach schemes to ministries by keyword; use `"jal shakti"` for water schemes.

**Why it was wrong.** There is no "Ministry of Jal Shakti" row in the published budget — the money sits under *Department of Drinking Water and Sanitation*. Two schemes silently ended up with zero allocations, and a gold query returned NULL.

**Replaced by.** Repointed to the real department name. **Shape B:** the code was not broken, it was matching against an assumption about the data rather than the data.

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## 30. Generating Revised Estimates for every year — *wrong, and it hid for a long time*

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**Decision.** `fiscal_years.re_available` marks the newest year as having no RE. Separately, the loader generated `re_amount_crore` for every year.

**Why it was wrong.** The flag and the data disagreed. Two evaluation questions asserted "no Revised Estimate exists for 2023-24" while the column held **1,167 values**.

**How it survived.** `verify_gold.py` reported **60/60 agreement** between the refusal rules and the gold set — and both were wrong in exactly the same way, because both were written from the same assumption. **This is Shape A in its purest form.** Checking each against *the data* found it in seconds.

**Replaced by.** Both the flag and every RE column now derive from one `re_published()` function, so they cannot drift apart.

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## 31. Counting "wrong order" as confidently wrong — *wrong, and it flattered my own number*

---

**Decision.** Any result not matching gold counted as confidently wrong.

**Why it was wrong.** Right rows in a different sequence has not misled anyone about a figure. Counting it inflated the headline in the project's favour — the one direction an evaluation cannot afford to err.

**Replaced by.** A separate `WRONG_ORDER` outcome, excluded from the headline and from `CORRECT`.

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## 32. Hardcoding a spread in the demo — *wrong*

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**Decision.** The demo script printed "spanning a 5x range".

**Why it was wrong.** The actual range was 1.8x. A demo for a project about unverified numbers was asserting an unverified number.

**Replaced by.** The demo computes the spread from live query results, and additionally checks whether the baseline's answer matches *any* certified reading — it did not; it had produced a fifth number.

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## 33. A semantic layer that only surfaced part of itself — *wrong*

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**Decision.** Render the semantic layer into the prompt.

**Why it was wrong.** `semantic_context()` surfaced only five of the eight documented ambiguities. Gross/net, borrowings and devolution were documented in the glossary and never reached the model. Three trap failures traced to exactly that gap.

**Replaced by.** Expanded coverage, plus column value-domains read **live from the database** so they cannot go stale, plus an explicit NULL-semantics section.

**Follow-up finding.** Expanding it did **not** move the headline. It fixed two specific behaviours and the net was zero, inside the noise. Reported as a behavioural improvement with no measured effect — **Shape C**.

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## 34. Framing refusal as a binary — *wrong*

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**Decision.** A question is either answered or refused.

**Why it was wrong.** The failure being measured is a *confident* wrong number. A figure delivered with its assumption stated is not confident, so it is not that failure. The binary framing missed a third option.

**Replaced by.** A switchable policy: STRICT refuses; DISCLOSE answers with a certified default and names the assumption. Both measured. A dimension only gets a default where convention supplies one — `estimate_basis` does, `scheme_identity` does not, because Swachh Bharat Gramin and Urban are two different schemes and picking one silently is the error, not the fix.

**Result:** refusal rate 46.7% → 30.0%, confidently wrong 16.7% → 21.7%. A real trade, honestly reported, with STRICT kept as the default.

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## 35. Trusting a read-only connection as a sandbox — *wrong, and security-relevant*

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**Decision.** A read-only DuckDB connection plus SELECT-only parsing is sufficient isolation.

**Why it was wrong.** A read-only connection blocks *writes* and nothing else. This is a pure SELECT that passes every structural check:

```
SELECT * FROM read_csv_auto('C:/Users/me/secrets.csv')
```

Confirmed by planting a value in a file outside the database and reading it back. **A language model writes this SQL** — a model that is confused, or steered by text in the schema it was shown, can emit one of these with nothing upstream noticing.

**Replaced by.** A block on 25 filesystem and external-catalog functions.

**The fix was then wrong twice, and only testing caught it:**

1. sqlglot models some of these as *typed* nodes ( `read_parquet` → `exp.ReadParquet` ), so an `Anonymous`-only check missed them.
2. On a typed node, `.name` returns the **first argument** — so `read_csv('x.csv')` reported itself as `"x.csv"` and slipped through again.

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## 36. My own audit script contained the bug it was auditing — *wrong*

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While writing independent cross-checks for the gold queries, five flagged. All five were bugs in the **audit**, not in gold. The most instructive:

- A cross-check rewrote `effective_to_year >= 2016` as `NOT (effective_to_year < 2016)`. In SQL, `NOT (NULL < 2016)` evaluates to `NULL`, not `TRUE` — silently dropping 102 of 107 rows. **The audit for a project about SQL traps fell into a SQL trap.**
- A check asserted no special-category state exceeds 20M population. Assam is 31.2M. The prior was simply wrong about the world.
- A tolerance of `1e-9` was tighter than accumulated floating-point rounding across ~1,000 rows can deliver.

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## 37. An adjudication script decoupled from its own input — *wrong*

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**Decision.** `adjudicate.py` loads `eval/questions.yaml`, then checks hardcoded lists.

**Why it was wrong.** It loaded the file and **never used it**. Renaming or adding a question would leave it reporting "12/12 verified" against a stale list forever. **Shape A again**, in a script written specifically to break Shape A.

**Replaced by.** It reconciles its lists against the YAML and refuses to run on drift. Verified by simulating a rename.

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## 38. Prose that drifted from the data — *wrong, twice*

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The README quoted a fan-out factor of 42×, then 35×, while the live value was 33×. The seed is fixed, but any change to *how many random draws occur* shifts the whole downstream stream, so generated figures move between revisions.

Separately, `variance.json` was overwritten by a later run and the README kept quoting the earlier figures — baseline 76.7% with 0.0 spread when the file said 76.1% with 5.0. The effect/noise ratio was being reported as 35.7× when it was 11.9×.

**Replaced by.** Headline figures injected from the results files by script, plus a caveat that generated figures move between revisions and `verify_gold.py`'s live output is authoritative over prose.

## Part VII — Measurement and results

### 39. The evaluation set

60 questions in four categories:

| CATEGORY     | COUNT | CORRECT BEHAVIOUR                                         |
|--------------|-------|-----------------------------------------------------------|
| Answerable   | 20    | answer, matching a hand-written gold query                |
| Trap         | 12    | answer, resisting a tempting wrong path that runs cleanly |
| Ambiguous    | 16    | <b>refuse</b> — more than one certified reading           |
| Unanswerable | 12    | <b>refuse</b> — no reading has data                       |

32 questions carry gold SQL. All 32 were independently re-verified by a *different* route — an alternate join path, an algebraic identity, or a known generation invariant — rather than by re-reading them.

### 40. The headline result

Same model, same database, same schema text. The only difference is whether `semantic/` is loaded.

| METRIC                   | BASELINE | WITH LAYER   | CHANGE    |
|--------------------------|----------|--------------|-----------|
| <b>Confidently wrong</b> | 76.1%    | <b>16.7%</b> | −59.4 pts |
| Execution accuracy       | 32.3%    | 53.1%        | +20.8 pts |
| Refusal recall           | 0%       | 100%         | +100 pts  |

Both are means of three runs.

### 41. Is that number real?

A headline quoted without its spread is not a falsifiable claim. Each arm was run three times at temperature 0.

|                                    | BASELINE               | WITH LAYER             |
|------------------------------------|------------------------|------------------------|
| Confidently wrong                  | 76.1% mean, 73.3–78.3% | 16.7% <b>every run</b> |
| Spread                             | 5.0 pts                | <b>0.0 pts</b>         |
| Questions with an unstable outcome | 3 of 60                | <b>0 of 60</b>         |

**Effect 59.4 points against noise 5.0 points — a ratio of 11.9×.**

The asymmetry is itself a finding: deterministic pre-generation refusal makes the semantic arm perfectly stable, while the baseline swings across five points because local inference is not bit-reproducible across server restarts.

**The rule this establishes:** any claimed improvement smaller than about 5 points is not distinguishable from noise. That rule has already caught one false positive in this project (§33).

## 42. Two models, one harness

|                                | QWEN2.5-CODER:7B | LLAMA3.1:8B  |
|--------------------------------|------------------|--------------|
| Confidently wrong, baseline    | 76.7%            | <b>76.7%</b> |
| Confidently wrong, with layer  | <b>16.7%</b>     | 35.0%        |
| Execution accuracy, with layer | 53.1%            | 18.8%        |
| Refusal recall                 | 100%             | 100%         |

**The most interesting result in the project.** The two baselines are *identical*. Swapping a general model for a code model is worth 34 points of execution accuracy once the layer is on, and moves the baseline confidently-wrong rate by **exactly nothing**.

The confidently-wrong rate is a property of **the schema's ambiguity**, not of the model's competence. A better model does not rescue you from it.

But the layer does not close the whole gap — after it, the two models are still 18 points apart. So the honest version of "fix the context, not the model" is narrower than the slogan: **context fixes the ambiguity half; model capability still owns the correctness half**.

## 43. Policy comparison

|                                   | BASELINE | STRICT       | DISCLOSE     |
|-----------------------------------|----------|--------------|--------------|
| Confidently wrong                 | 75.0%    | <b>16.7%</b> | 21.7%        |
| Refusal rate                      | 0%       | 46.7%        | <b>30.0%</b> |
| Answered with a stated assumption | 0        | 0            | <b>8</b>     |

DISCLOSE answers 10 questions STRICT refuses; 8 match the certified default exactly. The refusal rate falls 16.7 points — a system that refuses half of what it is asked stops being asked.

The cost is a new failure mode STRICT does not have: **the stated assumption and the computed figure can disagree**. A mislabel is worse than a refusal, because the label makes it look checked. STRICT remains the default.

# Part VIII — Running and extending

## 44. Running it

```
pip install duckdb sqlglot pandas pyyaml
```

```
python scripts/load_data.py      # build the database, deterministic from one seed
python scripts/verify_gold.py    # gold queries, refusal rules, all six traps
python scripts/run_eval.py       # both arms, prints the ablation
```

Generation needs a local Ollama server with a model pulled. `qwen2.5-coder:7b` fits in about 6 GB of VRAM.

```
python scripts/run_eval.py --arm all --model llama3.1:8b
python scripts/measure_variance.py --repeats 5
python scripts/demo.py
python scripts/ask.py "What was the Budget Estimate for the Ministry of Railways in 2019-20?"
```

---

## 45. Extending it

**Add a question.** Append to `eval/questions.yaml` with a category and, if answerable, gold SQL. Then run `verify_gold.py` (does it execute?), `audit_gold.py` (does an independent route agree?) and `adjudicate.py` (does it reconcile?). The last will refuse to run until its lists are updated — that is deliberate.

**Add an ambiguity dimension.** Add it to `semantic/metrics.yaml`, add a detection rule in `ambiguity.py`, and decide whether it belongs in `CERTIFIED_DEFAULTS`. It belongs there only if convention supplies an answer; if the alternatives are different *objects* rather than different framings of one object, it must not have a default.

**Swap the model.** `--model <tag>` for any Ollama model. For a non-Ollama provider, implement the `Provider` protocol in `generate.py` — one method, `complete(prompt, temperature)`.

**Change the refusal policy.** `--arm disclose`. The behaviour is defined by `CERTIFIED_DEFAULTS` in `ambiguity.py` and the pin text in `pipeline.py`; those two must agree, or the disclosure and the figure diverge.

**Port to another database.** `db.py` is the only module that touches DuckDB. The read-only guard uses `sqlglot`, which supports many dialects — change `dialect="duckdb"` and the `_FILESYSTEM_FUNCTIONS` list, which is engine-specific.

---

## Part IX — Limitations

- **Row-level data is synthetic.** Entity names, entity count and every magnitude are real; the detail beneath them is generated. Do not describe this as "real government data" without that qualification.
- **100% refusal recall is against *this* question set.** A set written by someone else would score lower. The rules are keyword-based, which buys determinism and auditability at the cost of missing paraphrases a model would catch.
- **One SQL dialect.** DuckDB only. Multi-dialect compilation was cut for scope.
- **No row-level security.** Every query runs as the same read-only role. Real warehouses enforce per-user visibility, and benchmarks that ignore it overstate accuracy.
- **Reproducible per commit, not across commits.** A change to how many random draws occur shifts the whole downstream data stream. Trust `verify_gold.py`'s live output over any number in prose.
- **The 16 ambiguous categorisations rest on a measured threshold, not a proof.** Every one conceals a discrepancy of at least 1.14×; whether refusing is the right response at 1.14× is a product judgement.
- **DISCLOSE can mislabel.** The stated assumption and the computed figure can disagree (2 of 60 with a 7B generator). STRICT has no such failure mode.

# Glossary

| TERM                        | MEANING                                                                                                                   |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Ablation</b>             | An experiment that removes one component and holds everything else fixed, so any change is attributable to that component |
| <b>Abstention</b>           | A system declining to predict rather than guessing, trading coverage for accuracy                                         |
| <b>BE / RE / Actuals</b>    | Budget Estimate, Revised Estimate, Actuals — three different figures for the same year                                    |
| <b>Confidently wrong</b>    | A query that runs cleanly and returns a plausible but incorrect number. The failure this project measures                 |
| <b>Core / lakh</b>          | Indian units: 1 crore = 10,000,000; 1 lakh = 100,000; 1 crore = 100 lakh                                                  |
| <b>CTE</b>                  | Common Table Expression — a named intermediate result inside a query, written <code>WITH x AS (...)</code>                |
| <b>EXPLAIN</b>              | A SQL command that plans a query without running it; used here as a cheap validity gate                                   |
| <b>Execution accuracy</b>   | Share of questions where the generated query returned the right rows. The conventional single metric                      |
| <b>Fan-out</b>              | A join that multiplies rows because the tables are at different grains, inflating any sum                                 |
| <b>Gold query</b>           | A hand-written correct SQL query for an evaluation question, used as the reference answer                                 |
| <b>Grain</b>                | What one row of a table represents — here, one year versus one month                                                      |
| <b>Multiset</b>             | A collection that counts duplicates; used for row comparison so a lost <code>GROUP BY</code> cannot pass                  |
| <b>Quantisation</b>         | Compressing a model's weights to fewer bits so it fits in less memory, at some cost to quality                            |
| <b>Selective prediction</b> | Allowing a model to decline low-confidence cases; the ML framing of this project's refusal behaviour                      |
| <b>Semantic layer</b>       | A written, versioned description of what data means: column glossary, certified metrics, validated join paths             |
| <b>Text-to-SQL</b>          | Turning a natural-language question into a database query automatically                                                   |
| <b>Three-valued logic</b>   | SQL's TRUE / FALSE / NULL. <code>NULL &lt; 2016</code> is NULL, not FALSE — the cause of a real bug in §36                |

Compiled from the repository at commit `7175bc1`. All measurements produced by `scripts/run_eval.py` and `scripts/measure_variance.py`, stored in `results/`. Where a figure is a mean, its range is given alongside. Rendered to PDF by `scripts/build_report.py`.